



Mark Scheme (Provisional)

Summer 2021

Pearson International Advanced
Subsidiary Level
In Chemistry (WCH13)
Paper 01: Practical Skills in Chemistry I

Question number	Answer	Additional Guidance	Mark
1(a)(i)	An answer that makes reference to the following points: • identification of material • justification of use	(1) nichrome / nickel-chromium (alloy) / NiCr / platinum / Pt Do not award just “nickel” or “chromium” (1) inert / does not react / does not give a flame colour Allow does not react with HCl Allow high melting temperature / does not melt in the flame Ignore does not burn	2

Question number	Answer	Additional Guidance	Mark
1(a)(ii)	A description that makes reference to the following points: • use of hydrochloric acid • description of method • flame colour of sodium • flame colour of potassium	(1) Allow any mention of HCl(aq) e.g. cleaning the wire or mixing solid and acid or making a paste/solution Ignore dilute HCl instead of HCl(aq) (1) (wire then) dipped in solid / paste / solution and placed in (hot / roaring / colourless / blue-cone) (Bunsen) flame Allow salt / compound / substance / paste /sample / solution for 'solid' on / over / under / near / show / above for 'in' (1) yellow / gold / golden Allow orange Allow combinations of allowed colours such as yellow-orange Ignore persistent / bright (1) Lilac Do not award purple If both correct colours are given without assigning to the correct metal/compound award 1 mark out of the 2. If colours are reversed award 1 mark out of 2.	4

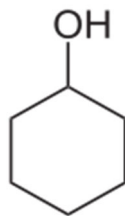
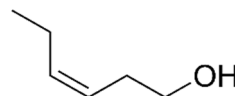
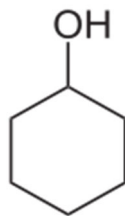
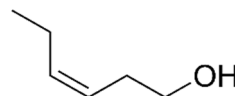
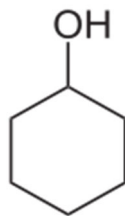
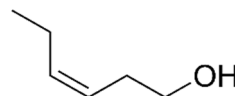
Question number	Answer	Additional Guidance	Mark
1(b)	A description that makes reference to the following points: • test for a sulfate • result of test for a sulfate • test for a carbonate • result of test for a carbonate	Tests can be in either order. Penalise unspecified acid once only in the two tests (1) barium chloride (solution) / $\text{BaCl}_2(\text{aq})$ and hydrochloric acid / $\text{HCl}(\text{aq})$ or barium nitrate (solution) / $\text{Ba}(\text{NO}_3)_2(\text{aq})$ and nitric acid / $\text{HNO}_3(\text{aq})$ Do not award sulfuric acid / H_2SO_4 (1) white precipitate Allow solid / suspension /ppt / ppte (1) addition of hydrochloric acid / $\text{HCl}(\text{aq})$ Allow any named acid (1) effervescence / fizzing / bubbles Allow gas given off which turns limewater cloudy Ignore just "gas"	4

(Total for Question 1 = 10 marks)

Question number	Answer	Additional Guidance	Mark
2(a)	An answer that makes reference to the following points: (add a few crystals / small amount of) phosphorus(V) chloride / phosphorus pentachloride / PCl_5 - steamy / misty fumes (of HCl)	(1) Allow thionyl chloride / SOCl_2 / Ester formation / reaction with Na Do not award PCl_3 (1) Allow white fumes Do not award white smoke Allow the formation of an ester M1 Addition of a carboxylic acid / named carboxylic acid and any identified strong acid Or Addition of an acyl chloride / named acyl chloride M2 Fruity / glue / characteristic smell of product Allow for addition of sodium M1 Addition of sodium / Na M2 formation of bubbles / fizzing / formation of a gas which burns with a squeaky pop Ignore just formation of gas / hydrogen	2

Question number	Answer	Additional Guidance	Mark
2(b)(i)	An answer that makes reference to one of the following pairs of points: • bromine water / aqueous bromine / bromine solution / bromine in organic solvent / Br₂(aq) • orange / yellow / brown to colourless OR • potassium manganate(VII) / KMnO₄ and sulfuric acid / H₂SO₄ • purple to colourless	In the results of both tests ignore “clear” (1) Allow bromine / Br₂(l) (1) Allow just decolourises Do not award red / red-brown with aqueous bromine but allow with bromine (1) Allow potassium permanganate Allow acidified potassium manganate(VII) (1) Allow just decolourises	2

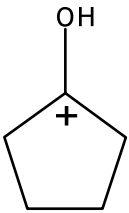
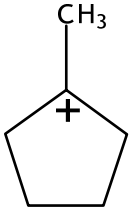
Question number	Answer	Additional Guidance	Mark
2(b)(ii)	An answer that makes reference to one of the following points: • orange / yellow / brown colour still present or • purple colour remains	Allow no change Allow does not decolourise Allow no reaction Allow TE on colours from b(i)	1

Question number	Answer	Additional Guidance	Mark								
2(c)	- any two correct - third correct	<table><tr><th>Compound</th><th>Colour change</th></tr><tr><td></td><td>orange to green</td></tr><tr><td></td><td>orange to green</td></tr><tr><td></td><td>no change / stays orange</td></tr></table> Award orange to blue three correct scores (2) any two correct scores (1) All three with the colours reversed (green to orange, green to orange and stays green) scores (1) Penalise the omission of the start colour once only	Compound	Colour change		orange to green		orange to green		no change / stays orange	2
Compound	Colour change										
	orange to green										
	orange to green										
	no change / stays orange										

Question number	Answer	Additional Guidance	Mark
2(d)(i)	An answer that makes reference to the following points: O-H bond and 3750–3200 (cm⁻¹)	In d(i) and d(ii) penalise a single value or a range within the range once only Allow 3200-3750 (cm ⁻¹) Do not award -OH / -O-H	1

Question number	Answer	Additional Guidance	Mark
2(d)(ii)	An answer that makes reference to the following points: C=C bond and 1669–1645 (cm⁻¹)	Allow 1669-1645 (cm ⁻¹) If no other mark awarded in (d)(i) and (d)(ii) allow (1) in (d)(ii) for identification of both bonds or both ranges	1

Question number	Answer	Additional Guidance	Mark
2(d)(iii)	An answer that makes reference to the following points: all three molecular ions have a $m/z = 100$ / all three compounds have a molar mass of 100 (g mol⁻¹)	Allow the three compounds are isomers Allow the compounds have the same molar mass Do not award the compounds have the same molecular ion	1

Question number	Answer	Additional Guidance	Mark
2(d)(iv)	An answer that makes reference to the following points:  - CH₃• / CH₃ (radical)	(1) Allow the + anywhere including outside of brackets around the structure Allow displayed formula $\overline{\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{COH}^+}$ Allow CH₂CH₂CH₂CH₂COH⁺ Ignore C₅H₉O⁺ / C₅H₈OH⁺ (1) Allow displayed formula Allow methyl (radical) Allow CH₃⁺ Allow (1) for  and OH• / OH / OH⁺	2

(Total for Question 2 = 12 marks)

Question number	Answer	Additional Guidance	Mark
3(a)	(from) yellow (1) (to) orange (1)	Correct colours reversed scores (1) Allow peach for orange Ignore modifiers e.g. pale Do not award to red or to pink	2

Question number	Answer	Additional Guidance	Mark								
3(b)(i)	two correct values	<table><tr><td>3</td><td>4</td></tr><tr><td>23.05</td><td>45.1(0)</td></tr><tr><td>1.25</td><td>23.20</td></tr><tr><td>21.8(0)</td><td>21.90</td></tr></table>	3	4	23.05	45.1(0)	1.25	23.20	21.8(0)	21.90	1
3	4										
23.05	45.1(0)										
1.25	23.20										
21.8(0)	21.90										

Question number	Answer	Additional Guidance	Mark
3(b)(ii)	An answer that makes reference to the following points: it is not concordant / not within 0.2 (cm³)	Allow not within 0.1 (cm³) Allow just 'it is rough / a trial / a rangefinder' Do not award uncertainty e.g. ± 0.1 / ± 0.2 (cm³)	1

Question number	Answer	Additional Guidance	Mark
3(b)(iii)	- calculation of mean titre (1) - calculation of moles of hydrochloric acid (1)	Example of calculation: $\frac{21.85 + 21.80 + 21.90}{3}$ = 21.85 (cm³) $\frac{21.85 \times 0.200}{1000}$ = 0.00437 / 4.37×10^{-3} (mol) Correct answer with some working scores (2) Allow TE on incorrect volumes in (b)(i) and on incorrect calculation of mean titre. Ignore SF except 1SF	2

Question number	Answer	Additional Guidance	Mark
3(b)(iv)	- calculation of moles of barium hydroxide in 10 cm³ (1) - calculation of moles of barium hydroxide in 1 dm³ (1) - calculation of concentration in g dm⁻³ to 2 or 3 SF (1)	Example of calculation: moles of Ba(OH)₂ in 10 cm³ = ans(b)(iii) / 2 = 0.002185 / 2.185 x 10⁻³ (mol) (ans(1)) moles of Ba(OH)₂ in 1 dm³ = ans(1) x 100 = 0.2185 / 2.185 x 10⁻¹ (mol) (ans(2)) = ans(2) x 171.3 = 37.429 g = 37 / 37.4 (g dm⁻³) to 2 or 3 SF Do not award 37.0 Correct answer with some working scores (3) Use of 171 gives 37.4 / 37.364 ALLOW TE throughout	3

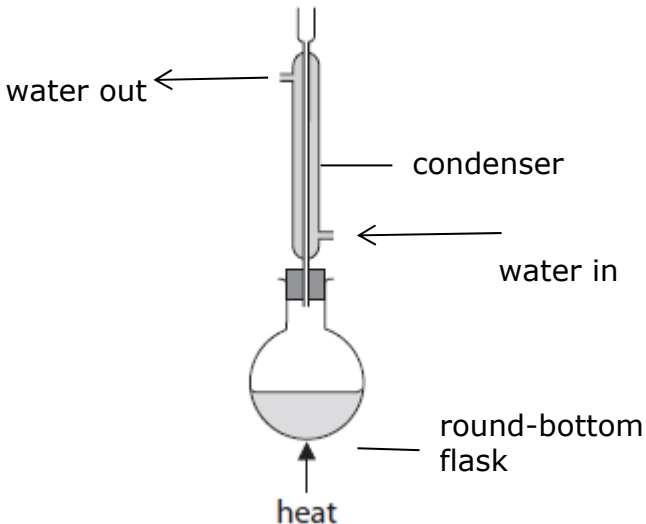
Question number	Answer	Additional Guidance	Mark
3(c)	An answer that makes reference to the following points: • use of a fume cupboard	Allow face mask Ignore laboratory coat / goggles / gloves / well ventilated laboratory Comment: Allow respiratory equipment	1

Question number	Answer	Additional Guidance	Mark
3(d)	An answer that makes reference to the following points: • oxidising (agent) / oxidiser / oxidant	Do not award flammable	1

(Total for Question 3 = 11 marks)

Question number	Answer	Additional Guidance	Mark
4(a)	An answer that makes reference to the following points: - gives a reason for cooling - gives a reason why this causes a reduction in yield	(1) reaction is exothermic / vigorous / gives off heat / Do not award Explosive Ignore violent / quick reaction (1) mixture may boil causing reactant / product to escape / reactant or product might evaporate or reaction may bubble / fizz / effervesce and overflow the round bottom flask (causing loss of reactant / product) Allow alkene formation / charring Ignore reference to equilibrium / rate / side reactions Do not award transfer losses	2

Question number	Answer	Additional Guidance	Mark
4(b)(i)	anti-bumping granules	Allow description of anti-bumping granules Allow other names: anti-bumping chips / beads boiling stones broken porcelain boiling chips	1

Question number	Answer	Additional Guidance	Mark
4(b)(ii)	A diagram that - contains a round bottom flask with contents and any indication of heating (1) - contains a vertical condenser with water jacket and correct water flow (1) - is a working apparatus: not stoppered, no gaps, a joint between flask and condenser (1)	Example of diagram:  Allow apparatus unlabelled Allow pear shaped flask Allow an arrow pointing upward as an indication of heat Ignore thermometers which do not seal the apparatus M1 is available for a distillation apparatus	3

Question number	Answer	Additional Guidance	Mark
4(c)(i)	An answer that makes reference to the following points: to neutralise (excess sulfuric) acid	Allow to remove (excess sulfuric) acid Allow hydrobromic acid / HBr / sodium hydrogensulfate / NaHSO₄	1

Question number	Answer	Additional Guidance	Mark
4(c)(ii)	An explanation that makes reference to the following points: - identifies the problem - gives a solution	(1) build-up of pressure (in the separating funnel) Ignore causes frothing Ignore formation of CO₂ (1) remove the stopper (with the funnel upright) or open the tap (with funnel inverted) Note: Remove the stopper to release the pressure would score 2. Comment: mark independently	2

Question number	Answer	Additional Guidance	Mark
4(c)(iii)	An answer that makes reference to the following points: drying agent / to remove water	Do not award dehydration	1

Question number	Answer	Additional Guidance	Mark
4(c)(iv)	An answer that makes reference to the following points: turns from cloudy to clear	Allow solution becomes clear Allow stops being cloudy Allow is no longer turbid	1

Question number	Answer	Additional Guidance	Mark
4(d)	An answer that makes reference to the following points: integer values of temperature in the ranges lower value 99 / 100 / 101(°C) and upper value 103 / 104 / 105(°C)	Award values to nearest 0.5 °C Do not award other fractions or decimals of degrees other than nearest 0.5 °C Do not award single values even if between the two ends of the range, e.g. 102 °C	1

Question number	Answer	Additional Guidance	Mark
4(e)(i)	reason for yield being less than 100%	Example of reasons: reaction incomplete transfer losses side reactions Ignore equilibrium reached Do not award the cold water bath was not used Do not award the procedure was not followed correctly Do not award spillages Do not award impure reagents	1

Question number	Answer	Additional Guidance	Mark
4(e)(ii)	• calculation of mass of 1-bromobutane required • calculation of moles of 1-bromobutane required • calculation of mass butan-1-ol required • calculation of percentage yield assumed OR • calculation of moles of butan-1-ol used • calculation of mass of 1-bromobutane for 100 % yield • calculation of volume of 1-bromobutane for 100 % yield • calculation of % yield if 20 cm³ prepared	Example of calculation: (1) = 20 x 1.27 = 25.4 (g) (1) = 25.4 ÷ 137 = 0.18540 (mol) (1) = 0.18540 x 74 = 13.720 (1) = 100 x 13.720 ÷ 21.0 = 65.332 (%) (1) = 21 ÷ 74 = 0.2837 mol (1) = 0.2837 x 137 = 38.878 g (1) = 38.878 ÷ 1.27 = 30.613 (cm³) (1) = 100 x 20 ÷ 30.613 = 65.332 (%) Do not award M4 if yield greater than 100% Use of 0.810 instead of 1.27 gives 41.699 (%) scores (3) Use of transposed M_r values gives 223.9 (%) scores (2) Allow use of 136.9 Allow TE throughout Ignore SF except 1SF Correct answer with some working scores (4)	4

(Total for Question 4 = 17 marks)

Total for Paper = 50 marks